

Off-Policy Policy Gradient

CS 185/285

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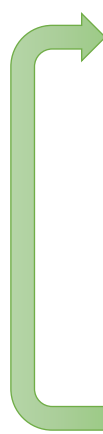
Part 1:

Back to policy gradients

Why do we *like* policy gradients?

a.k.a. on-policy batch actor-critic

policy gradient with GAE:

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\nabla_\theta J(\theta) \approx \sum_i \sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})$
 6. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

当优先考虑算法稳定性和可靠性，且对样本效率要求不高时，策略梯度是理想选择，比如机器人从仿真环境迁移到真实世界、大语言模型的强化学习训练。样本效率较低，需要更多的环境交互数据来优化策略，因此在样本获取成本高的场景中，可能需要结合其他方法（如离线强化学习）进行改进。

This makes policy gradients a great choice if we want a stable, reliable RL method, and we are not that concerned about sample efficiency

Particularly good for sim-to-real robotics and for training LLMs

policy gradient和policy iteration的区别：policy gradient是通过梯度来更新actor网络，可以平滑的更新，每次只改变一点
policy iteration每次都是使用argmax A function来更新，更新是跳跃的

Q-function methods can be hard to tune and might not converge

With on-policy methods, we can use Monte Carlo advantage estimators

$$\lambda = 1$$

$$\hat{A}_{\text{MC}}^\pi(\mathbf{s}_t, \mathbf{a}_t) = \sum_{t'=t}^{\infty} \gamma^{t'-t} r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) - \hat{V}_\phi^\pi(\mathbf{s}_t)$$

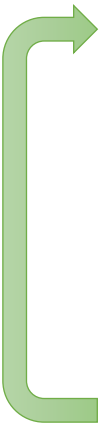
This gives us a provably **unbiased** policy gradient estimator

no matter how “wrong” the critic is!

The problem with policy gradients

a.k.a. on-policy batch actor-critic

policy gradient with GAE:

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\nabla_\theta J(\theta) \approx \sum_i \sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})$
 6. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

each iteration takes **one** gradient step

each iteration requires generating new samples

high variance means we usually want **bigger** batches than we would use with regular supervised learning, and smaller learning rates

this quickly becomes intractable

The kind of algorithm we would like to see

fictional “multi-step” policy gradient:

- 
- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
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 5. $\nabla_\theta J(\theta) \approx \sum_i \sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})$
 6. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

why is this not allowed?

$$\begin{aligned} & \mathbb{E}_{\tau \sim p_\theta(\mathbf{s}_t, \mathbf{a}_t)} [\nabla_\theta \log \pi_\theta(\mathbf{a}_t | \mathbf{s}_t) \hat{A}^\pi(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{\mathbf{s}_t, \mathbf{a}_t} p_\theta(\mathbf{s}_t, \mathbf{a}_t) \nabla_\theta \log \pi_\theta(\mathbf{a}_t | \mathbf{s}_t) \hat{A}^\pi(\mathbf{s}_t, \mathbf{a}_t) \end{aligned}$$

can we somehow *correct* this so
that we can take multiple steps?

Part 2:

Importance sampling

Off-policy learning & importance sampling

$$\theta^* = \arg \max_{\theta} J(\theta)$$

$$J(\theta) = E_{\tau \sim p_{\theta}(\tau)}[r(\tau)]$$

what if we don't have samples from $p_{\theta}(\tau)$?

(we have samples from some $\bar{p}(\tau)$ instead)

$$J(\theta) = E_{\tau \sim \bar{p}(\tau)} \left[\frac{p_{\theta}(\tau)}{\bar{p}(\tau)} r(\tau) \right]$$

$$p_{\theta}(\tau) = p(\mathbf{s}_1) \prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

$$\frac{p_{\theta}(\tau)}{\bar{p}(\tau)} = \frac{\cancel{p(\mathbf{s}_1)} \prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) \cancel{p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)}}{\cancel{p(\mathbf{s}_1)} \prod_{t=1}^H \bar{\pi}(\mathbf{a}_t | \mathbf{s}_t) \cancel{p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)}} = \frac{\prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)}{\prod_{t=1}^H \bar{\pi}(\mathbf{a}_t | \mathbf{s}_t)}$$

importance sampling

$$\begin{aligned} E_{x \sim p(x)}[f(x)] &= \int p(x) f(x) dx \\ &= \int \frac{q(x)}{q(x)} p(x) f(x) dx \\ &= \int q(x) \frac{p(x)}{q(x)} f(x) dx \\ &= E_{x \sim q(x)} \left[\frac{p(x)}{q(x)} f(x) \right] \end{aligned}$$

Deriving the policy gradient with IS

$$\theta^* = \arg \max_{\theta} J(\theta)$$

$$J(\theta) = E_{\tau \sim p_{\theta}(\tau)}[r(\tau)]$$

a convenient identity

$$p_{\theta}(\tau) \nabla_{\theta} \log p_{\theta}(\tau) = \nabla_{\theta} p_{\theta}(\tau)$$

can we estimate the value of some *new* parameters θ' ?

$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} r(\tau) \right]$$

the only bit that depends on θ'

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{\nabla_{\theta'} p_{\theta'}(\tau)}{p_{\theta}(\tau)} r(\tau) \right] = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{\cancel{p_{\theta'}(\tau)}}{\cancel{p_{\theta}(\tau)}} \nabla_{\theta'} \log p_{\theta'}(\tau) r(\tau) \right]$$

now estimate locally, at $\theta = \theta'$: $\nabla_{\theta} J(\theta) = E_{\tau \sim p_{\theta}(\tau)}[\nabla_{\theta} \log p_{\theta}(\tau) r(\tau)]$

The off-policy policy gradient

$$\theta^* = \arg \max_{\theta} J(\theta) \quad J(\theta) = E_{\tau \sim p_{\theta}(\tau)}[r(\tau)]$$

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \nabla_{\theta'} \log \pi_{\theta'}(\tau) r(\tau) \right] \quad \text{when } \theta \neq \theta'$$

$$\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} = \frac{\prod_{t=1}^H \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)}$$

$$= E_{\tau \sim p_{\theta}(\tau)} \left[\left(\prod_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \right) \left(\sum_{t=1}^H \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \right) \left(\sum_{t=1}^H r(\mathbf{s}_t, \mathbf{a}_t) \right) \right] \quad \text{what about causality?}$$

$$= E_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^H \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \underbrace{\left(\prod_{t'=1}^t \frac{\pi_{\theta'}(\mathbf{a}_{t'} | \mathbf{s}_{t'})}{\pi_{\theta}(\mathbf{a}_{t'} | \mathbf{s}_{t'})} \right)}_{\text{future actions don't affect current weight}} \left(\sum_{t'=t}^H r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \right) \left(\underbrace{\prod_{t''=t}^{t'} \frac{\pi_{\theta'}(\mathbf{a}_{t''} | \mathbf{s}_{t''})}{\pi_{\theta}(\mathbf{a}_{t''} | \mathbf{s}_{t''})}}_{\text{if we ignore this, we get a policy iteration algorithm (more on this next time)}} \right) \right]$$

future actions don't affect current weight

if we ignore this, we get
a policy iteration algorithm
(more on this next time)

A first-order approximation for IS

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^H \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \underbrace{\left(\prod_{t'=1}^t \frac{\pi_{\theta'}(\mathbf{a}_{t'} | \mathbf{s}_{t'})}{\pi_{\theta}(\mathbf{a}_{t'} | \mathbf{s}_{t'})} \right)}_{\text{exponential in } T} \left(\sum_{t'=t}^H r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \right) \right]$$

let's write the objective a bit differently...

on-policy policy gradient: $\nabla_{\theta} J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$

$(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) \sim \pi_{\theta}(\mathbf{s}_t, \mathbf{a}_t)$

off-policy policy gradient: $\nabla_{\theta'} J(\theta') \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})}{\pi_{\theta}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$

this is “a little bit wrong”
we'll see next time why

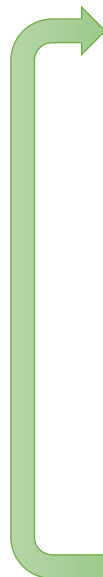
$$= \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \frac{\cancel{\pi_{\theta'}(\mathbf{s}_t^{(i)})}}{\cancel{\pi_{\theta}(\mathbf{s}_t^{(i)})}} \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$$

ignore this part

it's mostly ok to do this

The kind of algorithm we would like to see

actual “multi-step” policy gradient:

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\theta' \leftarrow \theta$
 - $K \times$ 6. $\nabla_{\theta'} J(\theta') \approx \sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$
 7. $\theta' \leftarrow \theta' + \alpha \nabla_{\theta'} J(\theta')$
6. $\theta \leftarrow \theta'$

Part 3:

Making importance sampling work

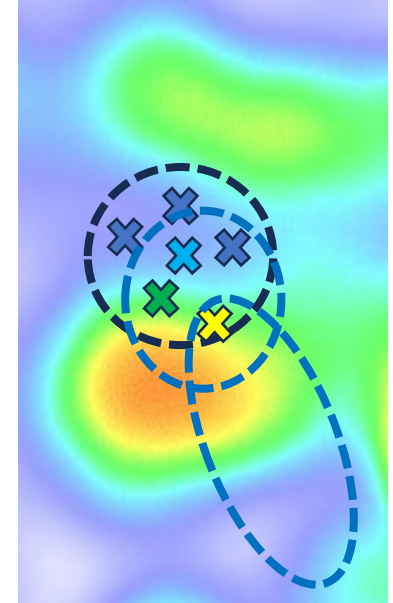
There is still a problem

actual “multi-step” policy gradient:

1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
5. $\theta' \leftarrow \theta$
- $K \times$ $\begin{aligned} &6. \nabla_{\theta'} J(\theta') \approx \sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)} \\ &7. \theta' \leftarrow \theta' + \alpha \underbrace{\nabla_{\theta'} J(\theta')} \end{aligned}$
6. $\theta \leftarrow \theta'$

once the weights are too far from 1.0,
the estimator is no longer useful

formally: its variance becomes too high



Clipped importance weights

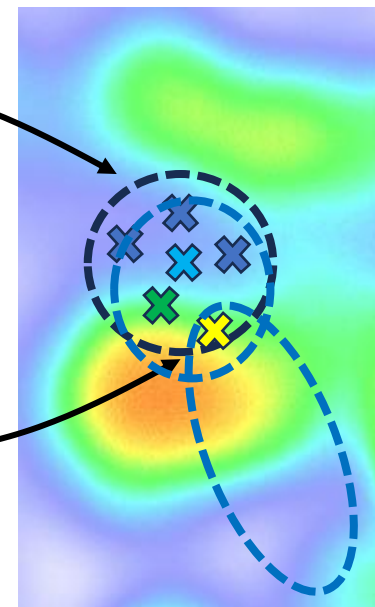
$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\overbrace{\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)}}^{w(\tau)} r(\tau) \right]$$

“surrogate objective”

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \nabla_{\theta'} \log p_{\theta'}(\tau) r(\tau) \right]$$

$w(\tau)$
keeps
decreasing here

$w(\tau)$
keeps
increasing here



try 0.1

↓

$$\text{redefine } w(\tau) = \max \left\{ 1 - \epsilon, \min \left\{ 1 + \epsilon, \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \right\} \right\}$$

puts a limit on how
much probabilities can
increase or decrease

One more detail

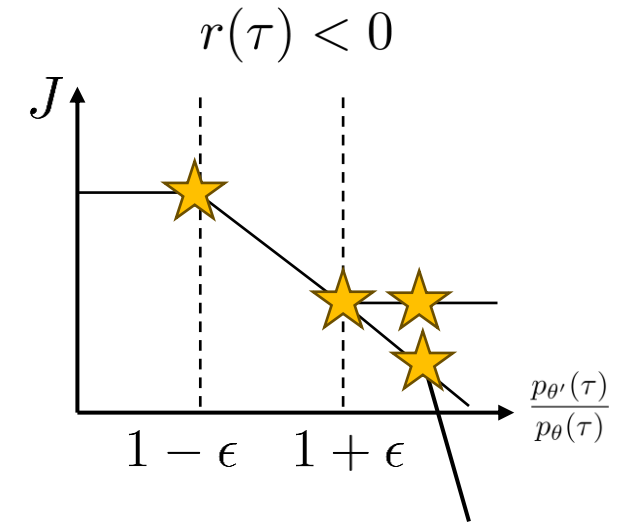
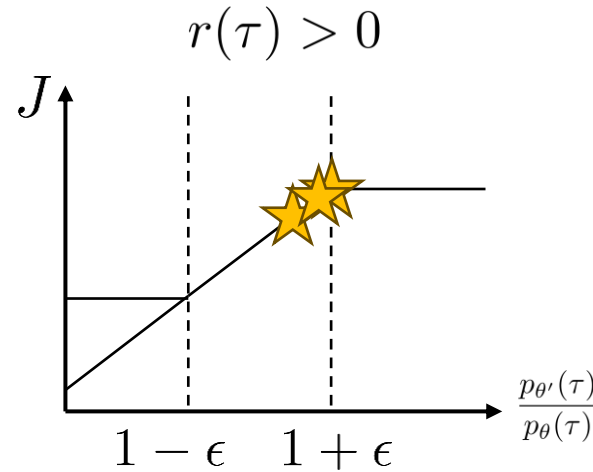
~~$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} [w(\tau)r(\tau)]$$~~

~~$$w(\tau) = \max \left\{ 1 - \epsilon, \min \left\{ 1 + \epsilon, \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \right\} \right\}$$~~

$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} [\min\{w(\tau)r(\tau), w_c(\tau)r(\tau)\}]$$

$$w(\tau) = \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)}$$

$$w_c(\tau) = \max \left\{ 1 - \epsilon, \min \left\{ 1 + \epsilon, \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \right\} \right\}$$



not paying any price
for drifting away from
data here!

Clipped importance sampling policy gradient

policy gradient with clipped IS (PPO):

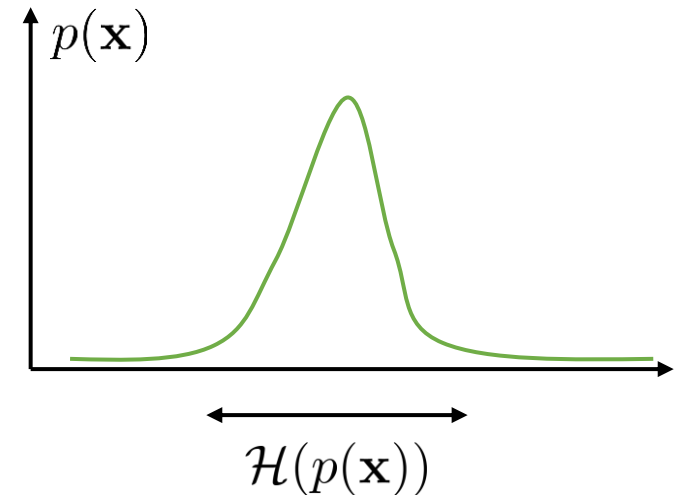
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
5. $\theta' \leftarrow \theta$
6. $\nabla_{\theta'} J(\theta') \approx \nabla_{\theta'} \mathcal{L}_{\text{CLIP}}(\theta')$
7. $\theta' \leftarrow \theta' + \alpha \nabla_{\theta'} J(\theta')$
6. $\theta \leftarrow \theta'$

this results in the widely used PPO algorithm

Some more details:

use GAE for $y_t^{(i)}$ as well

add $\sum_i \sum_t \mathcal{H}(\pi_{\theta'}(\cdot | \mathbf{s}_t^{(i)}))$ to objective
“entropy regularization”



$$\mathcal{L}_{\text{CLIP}}(\theta') = \sum_{i=1}^N \sum_{t=1}^H \min \left\{ \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} \hat{A}_t^{(i)}, \text{clip} \left(\frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t^{(i)} \right\}$$